

Algebra for Topology Review

Vincent Wang

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1 Introduction

Theorem 1.1 (Fundamental theorem of Homological Algebra). *Let $(C_*, \partial), (D_*, \partial)$ be free resolutions of M, N and $f : M \rightarrow N$. Then there exists $f_q : M_q \rightarrow N_q$ for all q such that the following diagram commutes:*

$$\begin{array}{ccccccc} M & \xleftarrow{\varepsilon} & C_0 & \xleftarrow{\partial} & C_1 & \xleftarrow{\partial} & \dots \\ \downarrow f & & \downarrow f_0 & & \downarrow f_1 & & \\ N & \xleftarrow{\varepsilon'} & D_0 & \xleftarrow{\partial} & D_1 & \xleftarrow{\partial} & \dots \end{array}$$

Further, any such f_* are chain homotopy equivalent.

Proof. We define the f_q inductively. First, we define f_0, f_1 . Since ε' is surjective and C_0 is free, there exists a map f_0 such that $\varepsilon' f_0 = f\varepsilon$. Since $\varepsilon' f_0 \partial = f\varepsilon \partial = f0 = 0$, so $\text{im } f_0 \partial \leq \ker \varepsilon' = \text{im } \partial$. Since C_1 is free, we can lift to f_1 such that $\partial f_1 = f_0 \partial$. Now suppose that the f_q are defined for $q \leq 1$ and that the diagram commutes for such q . Similarly, we have $\partial f_q \partial = f_{q-1} \partial^2 = f_{q-1} 0 = 0$, so $\text{im } f_q \partial \leq \text{im } \partial$, so lift to f_{q+1} such that $\partial f_{q+1} = f_q \partial$.

Now suppose f_*, f'_* are two such chain homotopies. We define $d_q : C_q \rightarrow D_{q+1}$ inductively as follows. First, we want $f'_0 - f_0 = \partial d_0$. We compute $\varepsilon'(f'_0 - f_0) = f\varepsilon - f\varepsilon = 0$, so because C_0 is free, we have $f'_0 - f_0 = \partial d_0$ for some d_0 . Now we wish for $f'_1 - f_1 = \partial d_1 + d_0 \partial$. We compute

$$\partial(f'_1 - f_1 - d_0 \partial) = (f'_0 - f_0) \partial - \partial d_0 \partial = \partial d_0 \partial - \partial d_0 \partial = 0$$

so write $f'_1 - f_1 = \partial d_1 + d_0 \partial$. Suppose d_q is defined for $q \leq 1$ and satisfies the chain homotopy condition. Then similarly:

$$\partial(f'_{q+1} - f_{q+1} - d_q \partial) = (f'_q - f_q) \partial - \partial d_q \partial = (\partial d_q + d_{q-1} \partial) \partial - \partial d_q \partial = 0.$$

□

Corollary 1.2. *Ext is well-defined.*

Proof. Let C_*, D_* be free resolutions of M and $f_* : C_* \rightarrow D_*, g_* : D_* \rightarrow C_*$ extend the identity 1_M . The chain $g_* f_*$ extends 1_M , so $g_* f_* \simeq 1_*$. Similarly, $f_* g_* \simeq 1_*$, so C_*, D_* are chain equivalent. Denote by C'_*, D'_* the cochain complexes formed by dualizing by $\text{Hom}(-, N)$. Then C'_*, D'_* are cochain equivalent, so Ext (being the cohomology of C'_*, D'_*) is well-defined. □

Lemma 1.3 (Horseshoe lemma). *Let $0 \rightarrow M' \xrightarrow{j} M \xrightarrow{p} M'' \rightarrow 0$ be short exact (not necessarily split) and C'_*, C''_* be free resolutions of M', M'' . Then there exists a free resolution C_* of M such that $0 \rightarrow C'_* \rightarrow C_* \rightarrow C''_* \rightarrow 0$ is short exact.*

Proof. Let $C_q = C'_q \oplus C''_q$. Since C''_0 is free and p is surjective, pick $k : C''_0 \rightarrow M$ such that $pk = \varepsilon''$. Define $\varepsilon(x'_0, x''_0) = j\varepsilon'x'_0 + kx''_0$. I claim that ε is surjective. Since ε'' is surjective, pick x''_0 such that $\varepsilon''x''_0 = px$, so $pkx''_0 = px$, so $x = kx''_0 + jx'$ for some $x' \in M'$ (by exactness at M). Since ε' is surjective, pick x'_0 such that $\varepsilon'x'_0 = x'$, so $x = \varepsilon(x'_0, x''_0)$.

Now we define the “error” maps $e_q : C''_q \rightarrow C''_{q-1}$ (for $q > 0$) to ensure the maps are compatible. We take $\partial_q(x'_q, x''_q) = (\partial'_q x'_q + e_q x''_q, \partial''_q x''_q)$. We wish for $\varepsilon\partial(x, y) = 0$, so

$$\varepsilon\partial(x, y) = \varepsilon(\partial'x + e_1 y, \partial''y) = j\varepsilon'(\partial'x + e_1 y) + k\partial''y = (j\varepsilon'e_1 + k\partial'')y$$

e_1 exists since $pk\partial'' = \varepsilon''\partial'' = 0$ and C''_1 is free, so we can pick a map $C''_1 \rightarrow M'$ such that when composed with j gives $k\partial''$. Since ε'' is surjective and C''_1 is free, we can pick e_1 . Suppose $\varepsilon(x, y) = j\varepsilon'x + ky = 0$. Applying p , we have $\varepsilon''y = 0$, so $y = \partial''z$. Then

$$\begin{aligned} j\varepsilon'x + k\partial''z &= j\varepsilon'x - j\varepsilon'e_1 z = j\varepsilon'(x - e_1 z) = 0 \\ &\rightarrow x - e_1 z = \partial'w \\ &\rightarrow (x, y) = \partial(w, z) \end{aligned}$$

Suppose e_q is defined for some $q \leq 1$. We wish for $\partial^2 = 0$, ie

$$\partial(\partial'x + e_{q+1}y, \partial''y) = (\partial'(\partial'x + e_{q+1}y) + e_q\partial''y, 0) = (\partial'e_{q+1}y + e_q\partial''y, 0)$$

so we wish for $\partial'e_{q+1} + e_q\partial'' = 0$. If $q = 1$, then $j\varepsilon'e_1\partial'' = -k\partial''^2 = 0$, so $\varepsilon'e_1\partial'' = 0$, giving the existence of e_2 . For $q > 1$, we have $\partial'e_q\partial'' = (-e_{q-1}\partial'')\partial'' = 0$, giving the existence of e_{q+1} .

Suppose $\partial(x, y) = 0$ (for some $q > 0$), so $\partial'x + e_q y = 0$ and $\partial''y = 0$. Pick z such that $y = \partial''z$, so $\partial'x + e_q y = \partial'x + e_q\partial''z = \partial'x - \partial'e_{q+1}z = \partial'(x - e_{q+1}z) = 0$, so let $x = \partial'w + e_{q+1}z$. Thus, $(x, y) = \partial(w, z)$.

It remains to show that the required squares commute. We compute $p\varepsilon(x, y) = \varepsilon''y$, $\varepsilon(x, 0) = j\varepsilon''x$, $p_{q-1}\partial_q(x, y) = \partial''_q p_q y$, and $\partial_q j_q x = \partial_q(x, 0) = (\partial'_q x, 0) = j_{q-1}\partial'_q x$. $\square$

Proposition 1.4. *If R is a commutative ring with identity such that every module over R is free, then R is a field.*

Proof. Suppose FTSOC $a \in R - 0$ such that $(a) \neq R$. The ring $R/(a) \neq 0$ is generated by one element and is free by assumption, so has rank 1. Thus, we have an isomorphism $R/(a) \rightarrow R$. Consider the composition $f : R \rightarrow R/(a) \rightarrow R$, where the first map is by projecting onto $R/(a)$. Thus, we have a surjective homomorphism of modules $f : R \rightarrow R$. f is determined by $f(1) = b$. Since f is surjective, there exists $c \in R$ such that $f(c) = f(c \cdot 1) = c \cdot f(1) = cb = 1$, so $b \in R^\times$, so f is also an isomorphism. It follows that the projection $R \rightarrow R/(a)$ is an isomorphism. However, $a \mapsto 0$, absurd. $\square$